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AI Climate Risk Intelligence Platform

Predicting Environmental Hazards with Machine Learning

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Date	12 June 2026
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1. Executive Summary

The AI Climate Risk Intelligence Platform is a machine-learning-powered web application designed to translate raw meteorological data into an instantly actionable Climate Hazard Risk Score (0–100). Developed as a capstone project at Presidency University, the platform ingests 533,312 real-world weather records, processes them through a fully automated ML pipeline, and serves risk predictions via a secure, role-based web dashboard.

The core predictive model — a Random Forest Regressor with 30 decision trees — achieved an R^2 accuracy of 89.27% on a cleaned dataset of 517,048 records. Seven meteorological parameters drive predictions, with Wind Speed alone contributing 85.6% of the model's predictive power. The platform surfaces risk in three tiers (Low / Moderate / High) and extends prediction to a 5-day forward forecast, making it suitable for emergency services, agriculture, and infrastructure management.

2. Problem Statement

Climate-related hazards are intensifying globally at an unprecedented rate. Emergency responders, agricultural planners, and infrastructure managers increasingly depend on real-time risk intelligence to make proactive, life-saving decisions. Yet the landscape of existing tools suffers from three critical gaps:

- **Complexity:** Existing forecasting tools are often too complex for non-specialist operators to interpret quickly.
- **Lack of AI:** Most tools lack AI-driven automation, relying on manual expert analysis that cannot scale.
- **Speed Gap:** The latency between raw data arrival and an actionable decision is too long for fast-moving hazard events.

This platform was built to close those gaps — delivering a single, normalized Hazard Index from raw weather inputs within seconds, accessible to both technical and non-technical users through a progressive web application.

3. Platform Overview

The platform is architected around five core capabilities, each designed to be self-contained yet tightly integrated:

Capability	Description
Massive Data Ingestion	Ingests 533,312 real meteorological records for robust model training.
Automated ML Comparison	Trains and benchmarks 3 ML models automatically, selecting the best performer.
Risk Score Output	Produces a Climate Hazard Risk Score (0–100) from 7 weather inputs in real time.
5-Day Forecast	Generates a multi-output forward forecast with predicted daily risk trends.
Accessible Interface	Delivered via a secure, role-based web dashboard and a mobile-ready PWA.

4. Dataset & Data Pipeline

The model is trained on a large-scale, real-world meteorological dataset. Raw records underwent a rigorous preprocessing pipeline before being used for training:

4.1 Data Processing Steps

Step	Detail
Raw Records	533,312 rows collected from meteorological sources
Duplicate Removal	16,264 duplicates removed → 517,048 clean records retained
Outlier Handling	IQR Capping (1.5× IQR rule) applied to clip extreme values
Feature Scaling	StandardScaler (Z-score normalization) applied to all 7 features
Training Sample	100,000 records randomly sampled; 80/20 train-test split

4.2 Input Features & Target Variable

Seven meteorological parameters form the model's input vector:

- Atmospheric Pressure (hPa)
- Global Solar Radiation (W/m²)
- Mean, Minimum, and Maximum Temperature (°C)
- Wind Speed (km/h)
- Wind Bearing (degrees)

The target variable is `normalized_label` — a continuous hazard index on a 0–1 scale, multiplied by 100 for presentation as the final Risk Score.

5. Risk Parameters & Feature Importance

Not all weather parameters contribute equally to hazard prediction. The table below shows each parameter's unit, its role in driving climate risk, and its computed importance in the trained Random Forest model:

Parameter	Unit	Role in Risk	Importance
Wind Speed	km/h	Primary driver — high winds = storm/cyclone risk	85.6%
Global Solar Radiation	W/m ²	Heat stress & drought risk indicator	9.4%
Min Temperature	°C	Frost & cold-wave hazard detection	2.0%
Max Temperature	°C	Heatwave & wildfire risk signal	1.4%
Atmospheric Pressure	hPa	Low pressure = storm system approaching	0.9%
Mean Temperature	°C	Overall thermal condition of the day	0.7%
Wind Bearing	degrees	Wind direction — affects storm path prediction	0.04%

Wind Speed's dominant 85.6% importance reflects its direct physical relationship to storm intensity. The remaining parameters, while smaller in isolated importance, are essential for detecting compound hazard scenarios such as simultaneous high-wind and high-temperature conditions.

5.1 Risk Score Classification

Score Range	Risk Level	Recommended Action
0 – 33	Low Risk	Monitor conditions; no immediate action required.
34 – 66	Moderate Risk	Alert relevant teams; review contingency protocols.
67 – 100	High Risk	Activate emergency response; issue public warnings.

6. The Algorithm — Random Forest Regressor

The platform's predictive engine is a Random Forest Regressor comprising 30 individual decision trees. The algorithm operates through the following end-to-end pipeline:

Random Forest was selected over alternative models (linear regression, gradient boosting) due to its robustness to outliers, ability to capture non-linear feature interactions, and interpretability via feature importance scores. The model's R^2 of 89.27% on the held-out test set confirms strong generalization to unseen meteorological conditions.

Stage	Description
1. Input Collection	7 weather parameters are collected for the target date/location.
2. Preprocessing	Inputs normalized via StandardScaler (Z-score) to equalize feature scales.
3. Bootstrap Sampling	Dataset randomly split into 30 subsets (one per tree), sampled with replacement.
4. Tree Building	Each subset trains an independent Decision Tree using random feature selection at each split node.
5. Individual Prediction	Each of the 30 trees independently outputs a risk score on a 0–1 scale.
6. Aggregation	All 30 tree predictions are averaged into a single ensemble output.
7. Output Scaling	Averaged value is multiplied by 100 to produce the final Hazard Index.
8. Classification	Score is categorized into Low / Moderate / High tiers with visual indicators.



7. Platform Features

The platform exposes its ML capabilities through a fully-featured, production-grade web application. Key modules include:

Feature	Details
Dashboard	Live KPIs and Chart.js-powered risk trend charts updated in real time.
Risk Calculator	7-parameter input form returning an animated 0–100 gauge with an AI-generated plain-language explanation.
5-Day Forecast	Multi-output Random Forest predicting both daily risk score and underlying weather parameters for each of the next 5 days.
Data Explorer	Paginated grid over all 517,048 clean records with search, sort, filter, and CSV/Excel export.
PDF Reports	Auto-generated risk assessment PDFs including GPU metadata for audit trails.
Security	JWT-based authentication with Admin and User role-based access control (RBAC).
Admin Panel	Trigger model retraining from the UI; view audit logs and GPU diagnostics (H200).
PWA Support	Offline-capable via Service Worker caching — accessible on mobile without an internet connection.

8. Real-World Applications

The platform's risk intelligence layer has immediate applicability across three sectors:

- **Agriculture:** Agricultural planning teams can use wind and heat-stress risk scores to schedule field operations, harvesting windows, and irrigation cycles, reducing crop losses from storm events.
- **Emergency Services:** Emergency services can integrate the 5-day forecast into dispatch planning and early-warning systems, enabling pre-positioned resources before high-risk events materialize.
- **Infrastructure:** Construction and infrastructure managers can enforce weather-related safety holds and adjust scheduling based on predicted High Risk periods, reducing accident risk and weather-related delays.

9. Current Limitations

The platform is a functional proof-of-concept with two primary constraints that bound its current scope:

- **Fixed Geography:** The training dataset is geographically fixed and does not yet generalize to global meteorological conditions. Predictions outside the source region may be unreliable.
- **Database Scale:** The backend uses a flat-file database architecture, which is not suitable for production-scale concurrent user loads or real-time streaming ingestion pipelines.

10. Future Work

The following enhancements are planned to evolve the platform toward a production-grade, globally deployable system:

- **Live Data Integration:** Integration with live weather APIs (OpenWeatherMap, NOAA) to replace static training data with real-time streaming meteorological feeds.
- **Advanced Models:** Replacement of the Random Forest model with sequence-aware architectures (LSTM or Transformer) to better capture temporal dependencies in weather patterns and improve multi-day forecast accuracy.
- **Cloud Deployment:** Cloud deployment on GCP or AWS to enable horizontal scaling, global availability, and sub-second inference latency under concurrent user load.
- **Push Alerts:** Mobile push notifications to deliver High Risk alerts to registered users on their smartphones, extending the platform's reach beyond the web dashboard.

11. Conclusion

The AI Climate Risk Intelligence Platform demonstrates that large-scale meteorological data, combined with a well-tuned ensemble ML model, can be transformed into real-time, actionable hazard intelligence accessible to non-specialist users. The Random Forest Regressor's 89.27% R^2 accuracy on 517,048 real climate records confirms the viability of this approach as a foundation for operational climate risk systems.

The platform's modular architecture — covering data ingestion, ML training, risk prediction, 5-day forecasting, and a secure web interface — provides a clear pathway to production deployment. With planned enhancements including live API feeds, LSTM-based temporal modelling, and cloud infrastructure, the system is positioned to evolve from a university capstone into a deployable tool for climate risk management at scale.

Key Result

Random Forest achieved 89.27% accuracy (R^2) on 517,048 real climate records — turning raw weather data into life-saving risk intelligence.